

BUSBY AGCM, AB, CanadaWMO: **712840**Lat: **53.9308N**Lon: **113.9217W**Elev: **714**StdP: **93.04**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.4	-28.3	-35.0	0.1	-31.4	-31.7	0.2	-28.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	27.5	16.7	25.8	16.1	24.3	15.5	18.7	24.6	17.5	23.6	16.5	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	12.8	21.9	15.2	11.8	20.2	14.2	11.0	19.0	56.1	24.4	52.2	23.8	48.9	22.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) N/A	N/A	N/A	DB	-35.8	30.5	2.6	1.8	-37.7	31.7	-39.2	32.8	-40.6	33.8	-42.5	35.0	(4)
(5)			WB	-35.9	20.8	2.6	1.5	-37.8	21.8	-39.3	22.7	-40.7	23.6	-42.5	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-10.5	-11.1	-5.4	3.1	10.5	14.3	16.5	15.4	10.5	3.7	-4.9	-12.2	(6)
(7)		DBStd	12.07	8.64	8.22	7.74	5.17	4.54	3.02	2.69	3.17	4.48	4.75	6.97	7.96	(7)
(8)		HDD10.0	3354	636	592	476	212	49	4	0	2	46	201	448	688	(8)
(9)		HDD18.3	5780	894	826	734	457	243	126	70	98	235	453	698	946	(9)
(10)		CDD10.0	643	0	0	0	6	66	132	201	171	62	6	0	0	(10)
(11)		CDD18.3	28	0	0	0	0	1	4	13	8	2	0	0	0	(11)
(12)		CDH23.3	542	0	0	0	1	50	76	191	159	63	2	0	0	(12)
(13)		CDH26.7	86	0	0	0	0	6	8	29	29	13	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	429	20	11	16	29	49	76	87	56	33	19	19	14	(15)
(16)		PrecMax	606	72	28	33	55	113	162	152	106	79	58	73	29	(16)
(17)		PrecMin	256	2	0	5	3	9	20	27	27	4	5	4	2	(17)
(18)		PrecStd	77	17	6	8	13	22	35	30	21	21	13	16	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.7	11.1	22.4	27.5	27.9	29.6	29.4	28.5	21.6	12.1	5.5	(19)
(20)			MCWB	2.9	2.8	5.7	10.7	13.4	16.1	18.4	17.0	16.0	12.2	7.1	1.8	(20)
(21)		2%	DB	5.2	4.0	8.0	17.8	24.6	25.6	27.2	26.9	25.0	16.3	7.6	3.2	(21)
(22)			MCWB	2.0	1.2	3.5	8.4	12.4	15.3	17.5	17.3	14.7	9.2	3.9	0.5	(22)
(23)		5%	DB	3.7	2.2	5.6	14.3	22.2	23.5	25.5	25.0	22.3	13.6	5.5	1.2	(23)
(24)			MCWB	1.1	0.0	2.3	6.5	11.4	14.3	17.1	16.6	13.7	7.5	2.5	-1.0	(24)
(25)		10%	DB	1.6	0.4	3.7	11.7	19.7	21.7	23.8	23.1	19.3	11.3	3.4	-1.2	(25)
(26)			MCWB	-0.3	-1.3	1.0	5.2	10.1	13.7	16.3	15.8	12.3	6.3	1.3	-2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.2	6.0	10.9	14.5	18.0	20.9	20.1	16.9	12.2	7.1	2.4	(27)
(28)			MCD B	6.2	6.2	10.3	21.8	24.4	24.5	26.7	24.8	25.8	20.6	10.9	5.0	(28)
(29)		2%	WB	2.3	1.6	4.0	8.6	13.2	16.4	19.2	18.6	15.4	9.8	4.5	0.6	(29)
(30)			MCD B	5.0	3.7	7.3	17.0	22.0	23.0	24.9	24.7	23.7	15.2	7.0	2.9	(30)
(31)		5%	WB	1.1	0.1	2.4	7.0	12.1	15.3	17.9	17.4	14.1	8.1	2.8	-1.0	(31)
(32)			MCD B	3.2	1.9	5.3	13.5	20.4	21.4	23.6	23.4	21.2	12.8	5.0	1.1	(32)
(33)		10%	WB	-0.3	-1.3	1.2	5.6	10.9	14.4	16.8	16.3	12.8	6.8	1.5	-2.9	(33)
(34)			MCD B	1.7	0.6	3.4	11.0	18.7	20.3	22.4	22.0	18.6	10.6	3.4	-1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	10.3	10.1	10.7	12.8	11.8	12.0	12.7	12.8	10.5	8.5	9.1	(35)
(36)			MCD BR	9.6	10.7	11.6	16.1	17.1	15.2	15.2	15.9	18.1	14.6	10.2	10.9	(36)
(37)			MCWBR	7.2	8.5	7.8	8.3	7.7	6.9	7.4	7.5	8.9	8.3	6.8	8.8	(37)
(38)		5% WB	MCD BR	9.6	10.4	10.9	14.9	15.4	13.0	13.6	14.0	17.0	13.8	9.2	10.9	(38)
(39)			MCWBR	7.5	8.3	7.5	8.1	7.3	6.3	7.5	7.4	8.7	8.2	6.7	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.268	0.303	0.324	0.340	0.350	0.353	0.357	0.307	0.280	0.251	0.236	(40)	
(41)		taud	2.310	2.336	2.325	2.393	2.394	2.379	2.420	2.354	2.500	2.551	2.454	2.313	(41)	
(42)		Ebn at Noon	736	844	881	908	907	898	888	861	874	832	746	682	(42)	
(43)		Edh at Noon	55	76	97	104	110	113	107	107	81	61	48	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.91	1.94	3.43	4.64	5.64	5.94	6.03	4.98	3.44	1.96	0.94	0.66	(44)	
(45)		RadStd	0.09	0.15	0.26	0.41	0.43	0.40	0.34	0.35	0.35	0.23	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page